

Simplify algebraic expressions



1

a $9u$

b $4a$

c 3

d 4

e $-35n$

f $\frac{3w}{8}$

g $21v$

h $6 + 5r$

i $-15v$

j $14v$

k $-13t$

l $\frac{16t}{27}$

m $8m^2$

n $-43n$

2

a 2

b $30t$

c $12m$

d $13v$

e $15m + 10$

f 3

g 6

h 2

i $9n^2$

j $-m$

k $18n^2$

l 3

3

a $34t$

b $27c$

c $5a$

d $36a$

e $8n^2$

f $27n^2$

g $22n^2$

h $13mn^2$

i $12w$

j $24pq$

k $4n$

l $12k^2$

m $8p^2$

n $10s$

o $2u$

p $6k^2$

Algebraic fractions

4

a 9

b $\frac{4w}{9}$

5

a 12

b $\frac{m}{6}$

6

a $\frac{3x}{2}$

b $-\frac{8y}{11}$

c $\frac{31y}{20}$

d $\frac{2x}{3}$

e $\frac{3x+2}{2}$

f $\frac{p}{4}$

g $\frac{17p}{35}$

h $\frac{11p}{15}$

i $\frac{5x+2}{12}$

j $\frac{7x+9}{6}$



Applications

7

a $(4 + 16n)$ cm

b $(14 + 2q)$ m

c $15b$ m

d $24v$ cm

e $24v$ cm

f $8a$ cm

g $(22n - 1)$ units

h $(27x + 10)$ units

i $(23d - 12)$ units

8

a $42v$ mm²

b $6q$ m²

c $50q^2$ cm²

d $25n^2$ cm²

e $36q^2$ m²

f $16n^2$ m²

9

a $56a^2$ mm³

b $(16a^2 + 84a)$ mm²

10 $\frac{2x + y + z}{2}$

11

a $\frac{3x - y}{9}$ dollars

b $\frac{3x - y + 11}{9}$ dollars

12 $120x^2$ cm²